Homework 5 — changed numbers (extra practice)

Modules 5–6 · Homework · No date

The numbers on this sheet are changed from her sheet.

Name _____ Date _____

1. A single 600mg dose of medication was administered orally. The medication is about 90% orally bioavailable and is characterized as having an absorption half-life of approximately 60 minutes, elimination half-life of approximately 5 hours, and apparent volume of distribution of 30 L.

1a Module 5

When does the maximum concentration of drug in the plasma occur?

1b Module 5

What is the expected maximum concentration following this single oral dose?

1c Module 5

What is the expected AUC following this single oral dose?

1d Module 5

If the dose were increased to 1200 mg, when does the maximum concentration of drug in the plasma occur?

1e Module 5

If the dose were increased to 1200 mg, what is the maximum concentration of drug in the plasma that you would expect?

1f Module 5

If the dose were increased to 1200 mg, what is the AUC that you would expect?

2. A 72 kg male patient received multiple IV bolus injections of 12 mg/kg every 6 hours for 36 hours. The medication has an elimination half-life of 3 hours and an apparent

volume of distribution that is 25% of body weight.

2a Module 6

What is the average concentration of drug in the plasma at steady state?

2b Module 6

What is the maximum concentration of drug in the plasma at steady state?

2c Module 6

What is the minimum concentration of drug in the plasma at steady state?

2d Module 6

What is the expected concentration of drug in the plasma 8 hours after administration of the last dose?